Olympic Insights

Using Data Analytics



Mohit

Contents:

- Objective
- Dataset and Tools
- Project Overview and Data Approach
- Data Model
- Page 1: Global Medal Overview
- Insights for page 1
- Page 2: Country Insights
- Insights for page 2
- Page 3: Athlete Spotlight
- Insights for page 3
- Page 4: Sport & Event Analysis
- Insights for page 4
- Page 5: Host Country Analysis
- Insights for page 5
- Page 6: Gender & Demographics
- Insights for page 6
- Limitations and Conclusions



Objective:



- To understand how the Olympic Games have grown and changed from 1896 to 2022, including the number of countries, athletes and events.
- To find out which countries win the most medals and how their performance has changed over the years.
- To find out which athletes and sports have won the most medals, and which athletes have played in the most Games.
- To check if a country wins more medals when it hosts the Games.
- To see how many men and women take part, and how the age of athletes has changed.
- To show all of this in one simple dashboard where anyone can filter by year, country or sport and see the answers quickly.

Dataset (6 tables):



Athlete Bio

name, sex, birth date, height, weight, country.

Athlete Event Results

Each athlete's events, sport and medal.

Country

country names and codes.

Games

year, city, host country and dates of each Olympics.

Medal Tally

Gold, Silver and Bronze count by country and Games.

Results

details of each event held at each Games.

Tools:

- Power BI (Power Query for cleaning, DAX for measures, visuals for dashboard).
- Navigation buttons and slicers for an interactive report.

Project Overview and Data Approach:



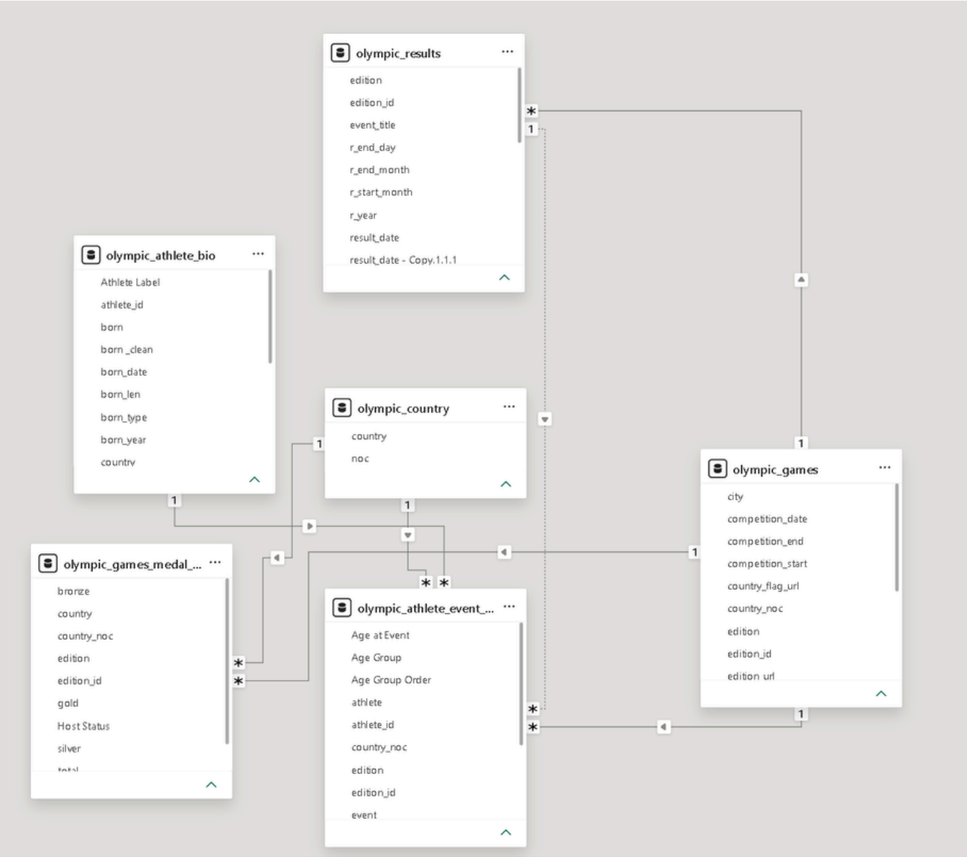
What we did

- Used 6 raw Olympic tables: athlete bio, athlete results, countries, games, medal tally and event results.
- Cleaned only what was needed using Power Query, then built a data model with relationships.
- Wrote DAX measures for KPIs and built 6 dashboard pages with navigation buttons.

Data cleaning steps

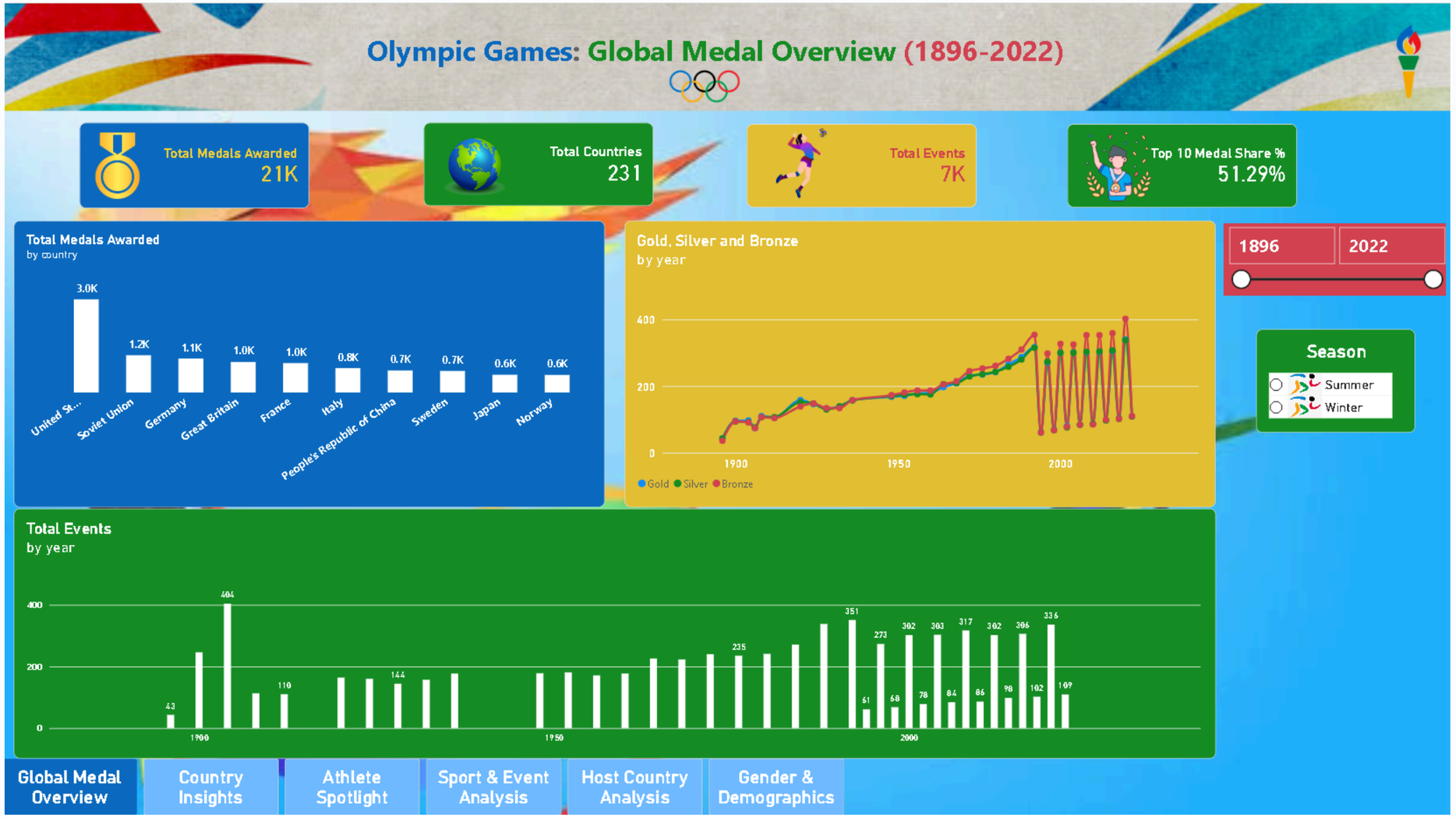
- Games dates: kept rows with missing dates (cancelled or no-data Games) and flagged them instead of deleting.
- Competition dates: split text like “14 May - 18 October” into proper start and end dates using the Games year.
- Athlete birth: split messy values into born_year, born_date and born_type, so full dates and year-only values are handled separately.
- Weight: converted ranges like “59-67” into a midpoint and added a flag column. Height and weight blanks were left as blank, not guessed.
- Results dates: removed time and notes, then extracted start and end dates.
- Country table: removed the duplicate ROC row so the relationship works.

Data Model:



- Games table connects to medal tally, results and athlete results using the edition column.
- Country table connects to medal tally and athlete results using the country code (noc).
- Athlete bio connects to athlete results using the athlete ID.
- This setup lets one slicer (year, country, sport) filter the whole page.

Page 1 – Global Medal Overview:



Insights for page 1 : Global Medal Overview



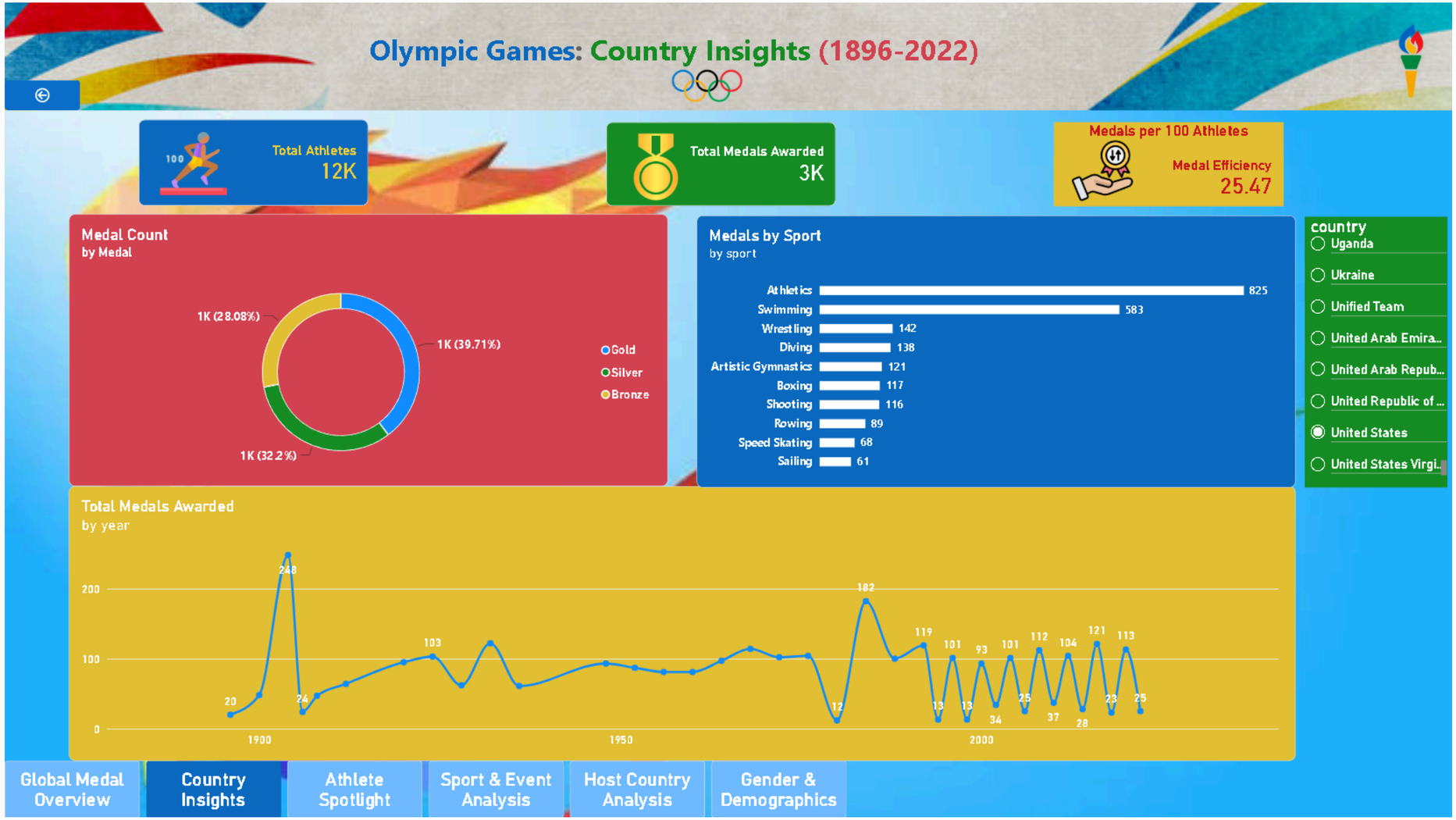
Approach

- Total Medals = Gold + Silver + Bronze from the medal tally, so team medals count once.
- Total Countries = distinct countries that took part, taken from athlete results.
- Top 10 countries chart uses a Top N filter on total medals.
- Year-wise charts use the Games year for medal trend and events held.

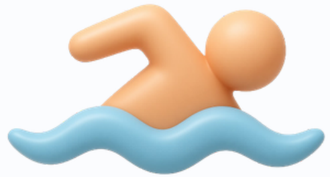
Insights

- The top 10 countries together hold about 51.29% of all medals. A few countries dominate the medal table.
- United States, Soviet Union and Germany lead the table with 3009 , 1204 and 1098 medals respectively.
- Medals and events grew steadily over the years, and the biggest jumps came around 1996 for medals and 1904 for total events.
- Breaks in the trend at 1916, 1940 and 1944 are because the Games were cancelled due to war.
- Participating countries grew from 31 in 1896 to 91 in 2022.

Page 2: Country Insights



Insights for page 2: Country Insights



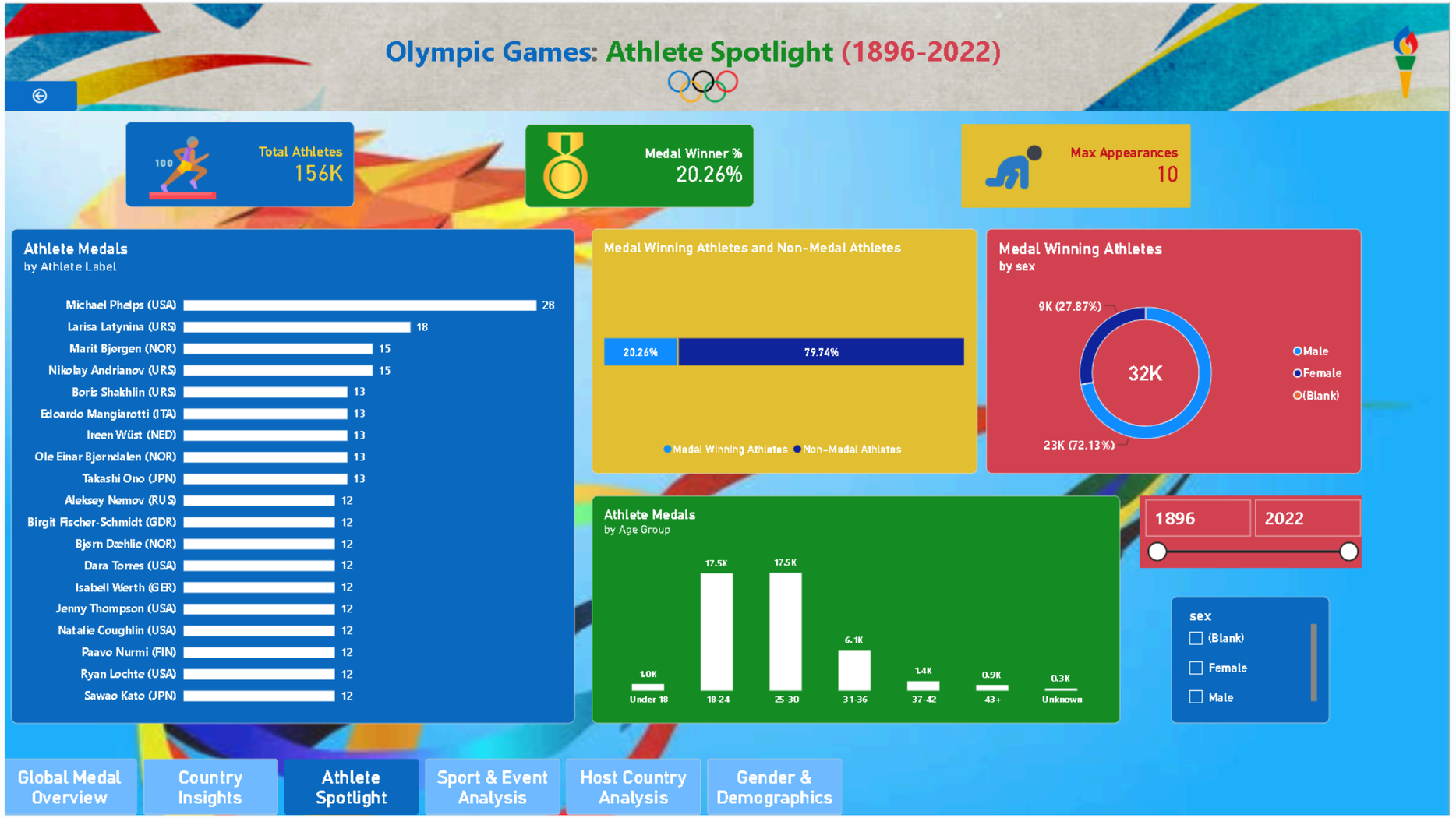
Approach

- One country slicer controls all visuals on the page.
- Medal Efficiency = (Total Medals ÷ Total Athletes) × 100, so we can compare countries of different sizes.
- Medal distribution shows Gold, Silver and Bronze for the chosen country.
- Best sports counts unique medals per sport, so a team medal counts once.

Insights

- Big countries win the most medals, but efficiency shows which countries convert athletes into medals better.
- Soviet union has an efficiency of 39.87 medals per 100 athletes, higher than United States.
- Most countries win medals in a small group of sports. United states's best sports are Athletics, Swimming and Wrestling.
- United states's medals rose in 1904 and dropped around 1906.
- Countries with very few athletes can show a high efficiency, so it should be read together with total athletes.

Page 3: Athlete Spotlight



Insights for page 3: Athlete Spotlight



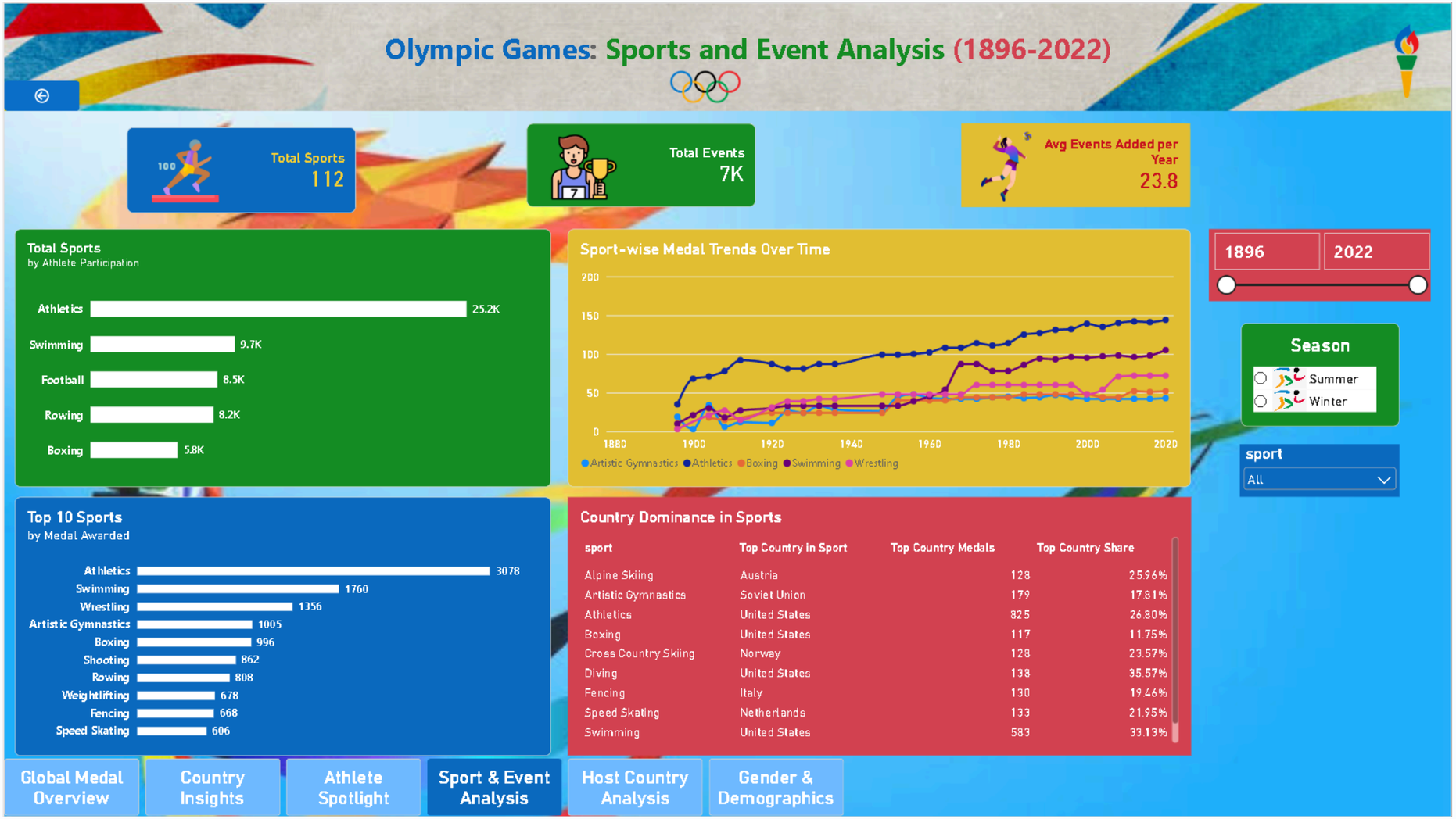
Approach

- Linked the athlete bio table to athlete results using the athlete ID.
- Counted medals per athlete, and counted distinct Games to measure Olympic appearances.
- Age at medal win = Games year minus birth year, then grouped into age bands.
- Gender distribution uses distinct medal-winning athletes by sex.

Insights

- A small group of athletes wins a large number of medals. Micheal Phelps leads with 28 medals.
- Only about 20.26% of all athletes ever won a medal, so most participants leave without one.
- The maximum number of Olympic appearances by any athlete is 10 Games.
- Most medals are won in the 25-30 age group.
- Among medal winners, 72.13% are male and 27.87% are female.

Page 4: Sport & Event Analysis



Insights for page 4: Sport & Event Analysis



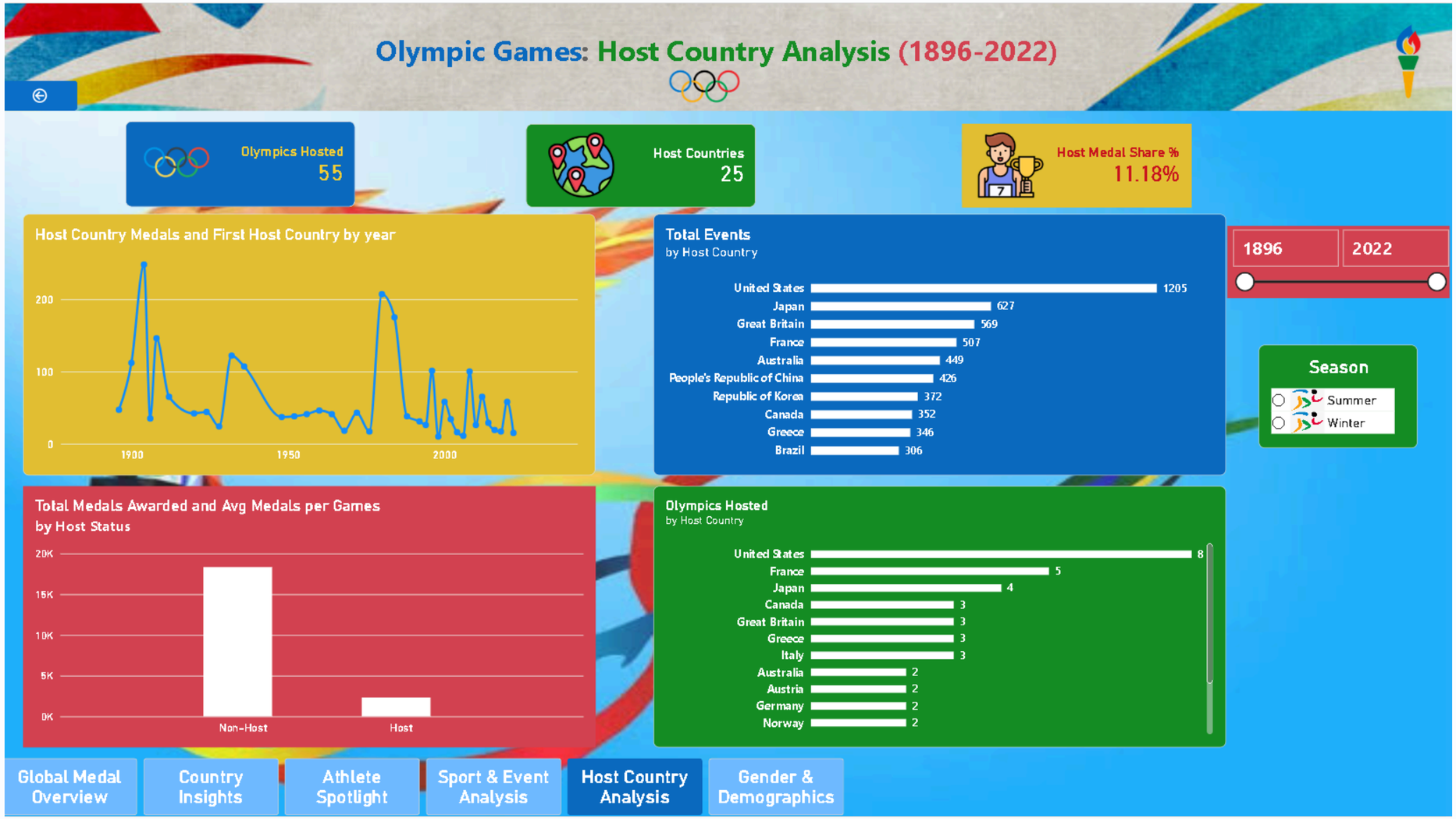
Approach

- All numbers come from the athlete results table, so the sport slicer works for the whole page.
- Medals Awarded counts each team medal once.
- Events Added uses the first year an event appeared in the Games.
- Country dominance finds the country with the most medals in each sport and its share.

Insights

- A few sports, such as Athletics and Swimming, give out a big share of all medals because they have many events.
- On average, 23.8 new events were added per Games Year.
- In some sports one country dominates, for example United States holds 26.80% of medals in Athletics.
- Athletics has the most athletes, so it is also the most competitive.
- Medals grew in all of the top 5 sports, most in Athletics and least in Artistic Gymnastics, because Athletics gained more new events. No sport declined.

Page 5: Host Country Analysis



Insights for page 5: Host Country Analysis



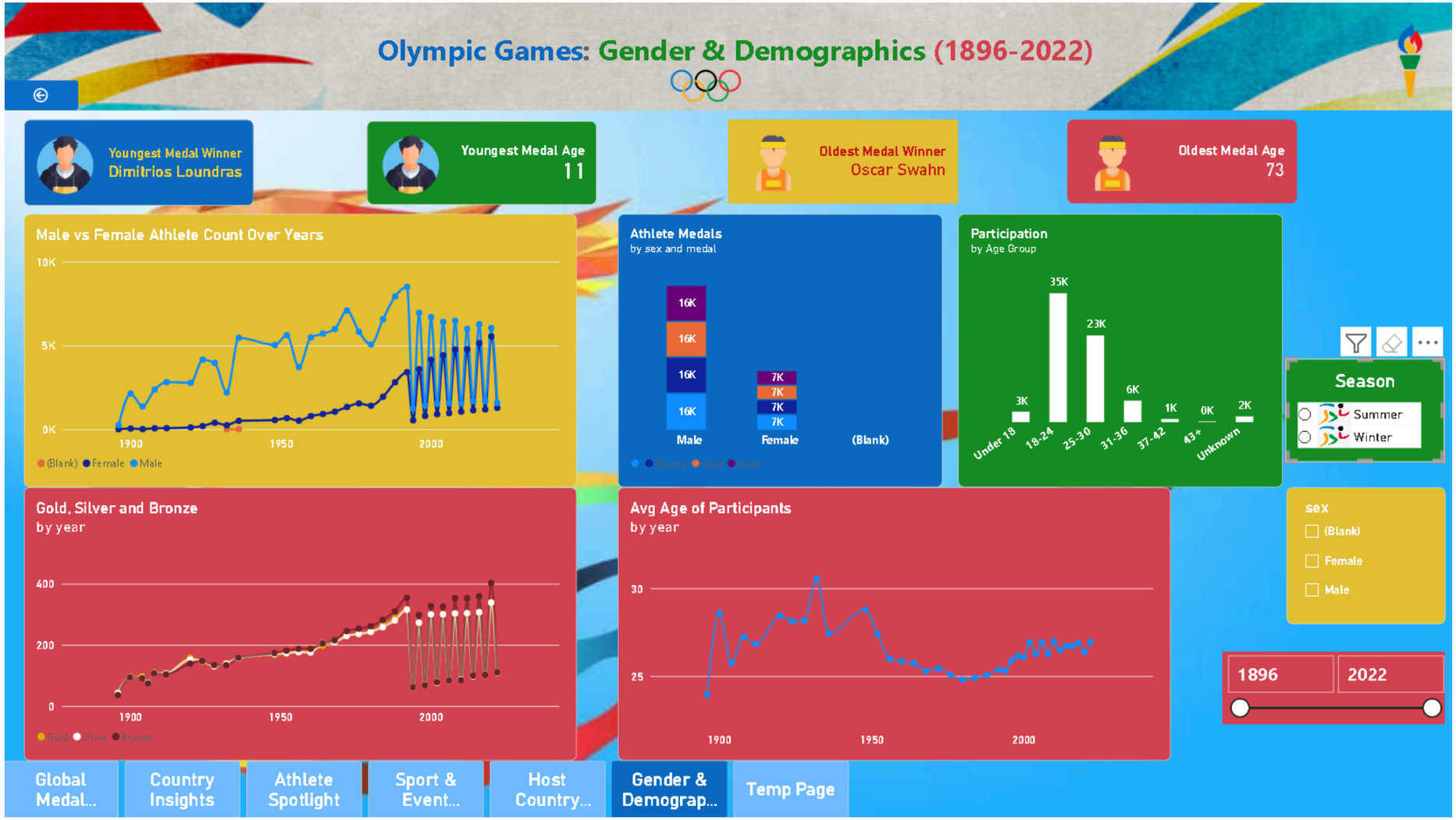
Approach

- Found each Games' host country from the Games table and tagged each medal row as Host or Non-Host.
- Counted only Games that actually have data, so cancelled Games are not counted as hosted.
- Compared medals as host vs non-host using an average per country per Games, to keep the comparison fair.
- Events conducted come from the results table, grouped by host country.

Insights

- United States hosted the Olympics the most times, with 8 Games.
- Host countries win about 4 times more medals (42 vs 10 on average) than they do as non-hosts.
- China's medals rose from 63 in 2004 to 100 in 2008, the year it hosted, and stayed high at 92 in 2012.This shows that the boost from hosting can last into the next Games.
- Summer Games hold more events than Winter Games. Apart from a one-time high of 404 events in 1904, the number of events held per Games has grown steadily since 1976.

Page 6: Gender & Demographics



Insights for page 6: Gender & Demographics



Approach

- Counted distinct athletes by sex for each year to see participation by gender.
- Gender-wise medals are split by Gold, Silver and Bronze.
- Youngest and oldest medal winners come from the minimum and maximum age at medal win.
- Average age is calculated once per athlete, so athletes who play many events do not count more than once.

Insights

- Women's participation has grown strongly over time, from 40 female athletes in 1900 to 5,539 in 2020. Participation has also been higher in Summer Games than in Winter Games..
- Women account for about 29% of medals won by athletes (6,526), while men account for about 71% (16,028).
- The average age of participants has stayed steady at around 27 years in recent Games, such as 2010, 2016 and 2022.
- Most participants fall in the 18-24 age group, with fewer in the youngest and oldest groups.
- The youngest medal winner was 11 years old and the oldest was 73. Both are rare cases from early Games.

Limitations and Conclusion:



Limitations

- The medal tally only has countries that won medals, so some averages, like the host vs non-host comparison, are slightly higher than reality.
- Ages are approximate because many athletes have only a birth year, not a full birth date.
- Some Games have no data (2024 and 2028) or were cancelled (1916, 1940, 1944), so they are left out of the analysis.
- Russia, ROC and the Soviet Union are separate entries in the data, so their medals are not combined.
- Athlete medal counts count each team member's medal, so they are higher than the official medal tally, which counts a team medal once.
- Hosts are usually strong sporting nations, so part of the host advantage may come from their normal strength.
- Some early records look unusual, such as 2 female athletes in 1896 and the large 1904 Games, and may reflect data quirks or very few participating countries.
-

Conclusion

- The dashboard shows how the Olympics grew in size, who wins, and who takes part.
- Medals are concentrated among a few countries, and host countries win about 4 times more medals than usual.
- Women's participation has grown strongly, and the average age of athletes has stayed steady at around 27 years in recent Games.
- It helps compare countries, sports, athletes and hosts in one place.


THANK
YOU